

Relativistic Jets in Active Galactic Nuclei: A Comparative Study of Blazars and M87

Mohammad Mahdi zare

Department of Physics

Student ID: 403100795

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Abstract

Relativistic jets are among the most energetic and visually striking phenomena in astrophysics, commonly observed in active galactic nuclei (AGN). Their observed properties depend strongly on special relativistic effects such as Doppler boosting, viewing angle, and time dilation. In this project, we develop a modular numerical framework to simulate relativistic jets using physically motivated but computationally simple models. We compare two representative cases: a blazar with a small viewing angle and the radio galaxy M87 with a larger inclination. Brightness profiles, synchrotron spectra with cooling breaks, three-dimensional jet geometry, and time-dependent lightcurves are computed and analyzed. The simulations demonstrate how a single intrinsic jet model can reproduce markedly different observational appearances solely due to relativistic effects.

1 Introduction and Motivation

Relativistic jets are highly collimated plasma outflows launched from the vicinity of supermassive black holes in active galactic nuclei (AGN). They transport energy and momentum over kiloparsec scales and radiate across the entire electromagnetic spectrum. Despite their complexity, many observational differences between AGN classes can be explained through geometry and special relativity.

Blazars are AGN whose jets are oriented close to the observer's line of sight. As a result, relativistic beaming strongly enhances their observed flux and compresses variability timescales. In contrast, radio galaxies such as M87 are observed at larger viewing angles, leading to weaker boosting and more extended jet structures.

The primary goal of this project is to demonstrate, through numerical simulations, how relativistic kinematics and radiation physics shape the observed properties of jets. Rather than constructing a fully magnetohydrodynamic model, we adopt simplified but physically transparent prescriptions that highlight the underlying mechanisms.

2 Relativistic Kinematics of Jets

2.1 Four-Vectors and Plasma Motion

In special relativity, the motion of particles in the jet is described using four-vectors. The four-velocity is defined as

$$u^\mu = \gamma(c, \vec{v}), \quad (1)$$

where $\vec{v}$ is the plasma velocity, c is the speed of light, and

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \quad (2)$$

is the Lorentz factor.

The corresponding four-momentum is

$$p^\mu = mu^\mu, \quad (3)$$

where m is the particle mass. This formalism allows a consistent relativistic description of energy and momentum transport along the jet.

2.2 Relativistic Beaming and Doppler Factor

Radiation emitted by relativistically moving plasma is strongly affected by Doppler boosting. The Doppler factor is defined as

$$\delta = \frac{1}{\gamma(1 - \beta \cos \theta)}, \quad (4)$$

where $\beta = v/c$ and θ is the angle between the jet direction and the observer.

The observed flux density transforms as

$$F_\nu^{\text{obs}} \propto \delta^{3+\alpha} F_\nu^{\text{em}}, \quad (5)$$

where α is the spectral index. This strong dependence on θ is the key reason why blazars appear dramatically brighter than radio galaxies.

2.3 Curved Jets

Real astrophysical jets are not perfectly straight. To model this effect, we allow the effective viewing angle to vary along the jet:

$$\theta(z) = \theta_0 + \epsilon z, \quad (6)$$

where θ_0 is the initial viewing angle and ϵ is a curvature parameter. This introduces spatial variations in Doppler boosting and directly affects the brightness profile.

3 Synchrotron Radiation and Cooling

Relativistic electrons spiraling in magnetic fields emit synchrotron radiation. For a power-law electron energy distribution,

$$N(E) \propto E^{-p}, \quad (7)$$

the emitted spectrum follows a power law,

$$F_\nu \propto \nu^{-\alpha}, \quad \alpha = \frac{p-1}{2}. \quad (8)$$

At high energies, radiative cooling becomes important and steepens the spectrum. This leads to a cooling break at frequency ν_{break} :

$$F_\nu \propto \begin{cases} \nu^{-\alpha_1}, & \nu < \nu_{\text{break}}, \\ \nu^{-\alpha_2}, & \nu > \nu_{\text{break}}, \end{cases} \quad \alpha_2 = \alpha_1 + 0.5. \quad (9)$$

4 Simulation Strategy and Implementation

The simulations were developed incrementally, with each stage adding a new layer of physical realism.

4.1 Stage I: One-Dimensional Jet and Brightness Profile

The jet is discretized along its length z . At each point, the local Doppler factor is computed and the emitted synchrotron spectrum is integrated over frequency. This yields the observed brightness as a function of distance along the jet.

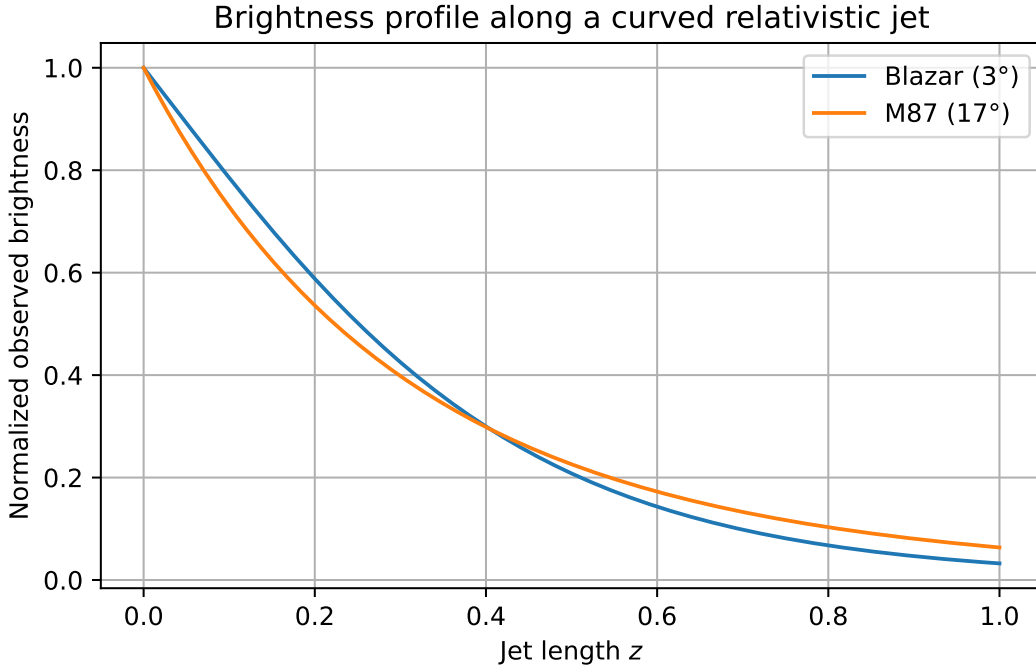


Figure 1: Normalized brightness profiles along the jet for a Blazar (3°) and M87 (17°). The Blazar emission is strongly concentrated near the jet base due to Doppler boosting.

4.2 Stage II: Spectral Modeling with Cooling Break

Using the same physical parameters, the observed synchrotron spectrum is computed. Only the viewing angle differs between the two sources.

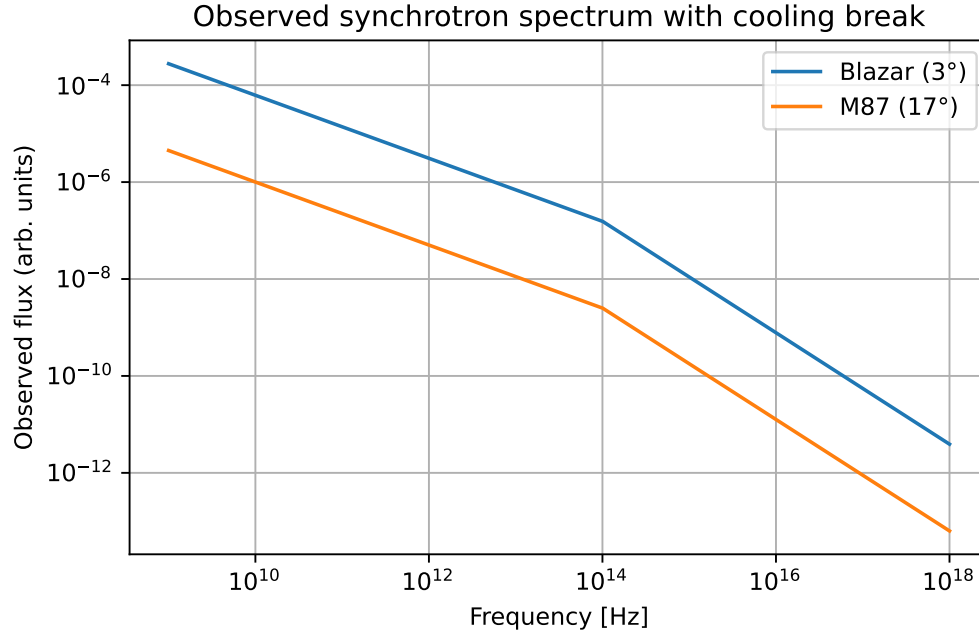


Figure 2: Observed synchrotron spectra including a cooling break. The overall normalization differs due to Doppler boosting, while the intrinsic spectral shape remains the same.

4.3 Dedicated Spectral Modeling of M87

In addition to the comparative spectral analysis, we performed a dedicated simulation of the synchrotron spectrum of M87 using parameters motivated by observations. Unlike blazars, M87 is observed at a relatively large viewing angle, resulting in weaker Doppler boosting and a more pronounced cooling signature.

The spectrum is computed using the same electron energy distribution as before, but with the viewing angle fixed at $\theta = 17^\circ$. This allows a clearer identification of the synchrotron cooling break and a more direct qualitative comparison with observational data.

4.4 Stage III: Three-Dimensional Jet Geometry

To visualize the effect of curvature, the jet is embedded in three dimensions. The brightness of each segment is encoded as a color scale.

4.5 Stage IV: Time-Dependent Lightcurve

Finally, slow jet precession is introduced, leading to time-dependent viewing angles. This produces variability in the observed flux.

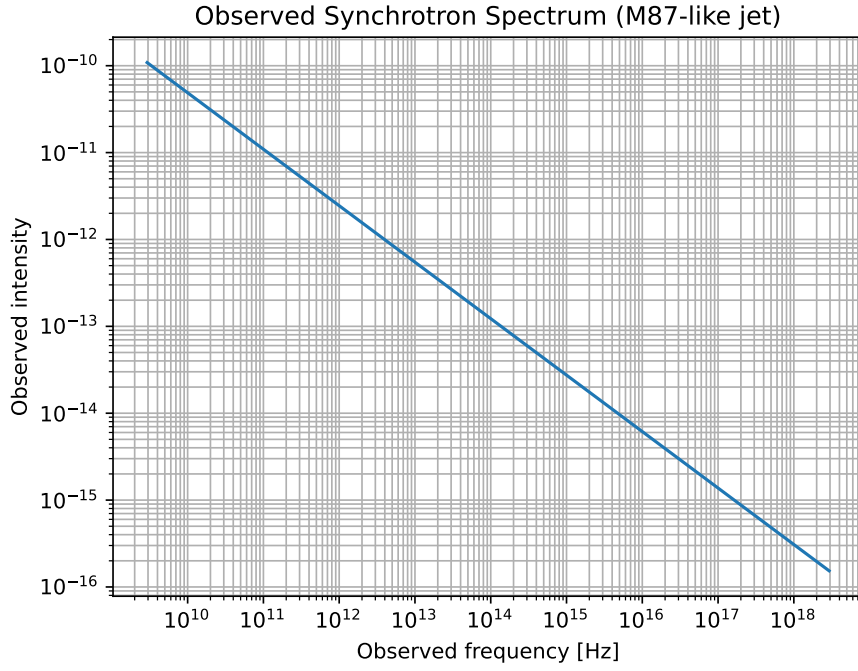


Figure 3: Simulated synchrotron spectrum of M87. The cooling break is clearly visible and the overall flux level is lower compared to the blazar case due to reduced Doppler boosting.

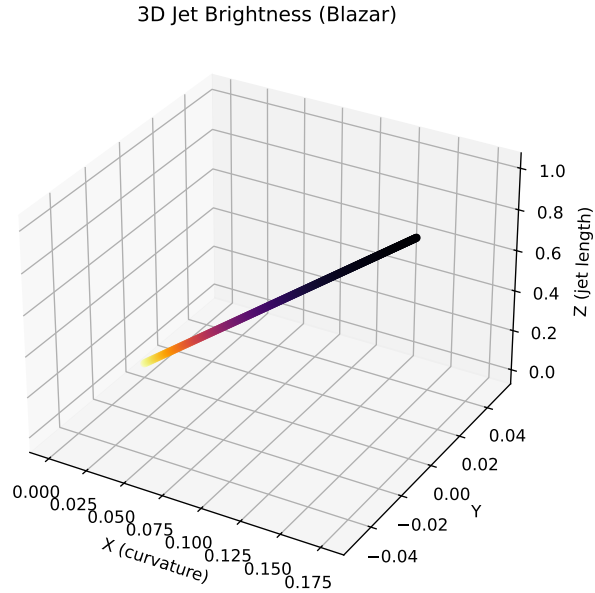


Figure 4: Three-dimensional visualization of a curved relativistic jet. Brightness variations arise from changes in the local Doppler factor.

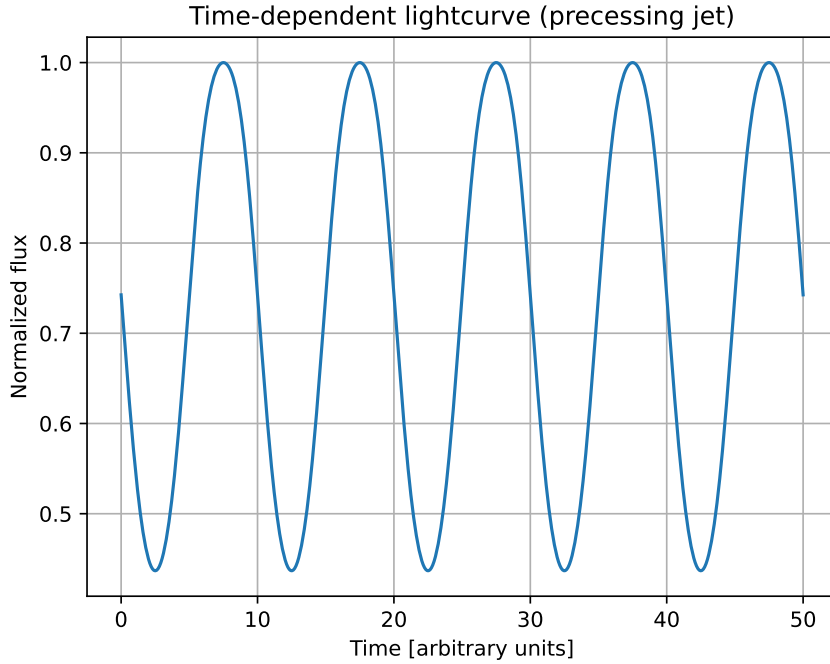


Figure 5: Simulated lightcurve of a precessing blazar jet. Small changes in viewing angle lead to strong flux variability.

5 Synchrotron Self-Compton (SSC) Emission

In addition to synchrotron radiation, relativistic electrons in AGN jets can further interact with the photons they have produced. This process, known as Synchrotron Self-Compton (SSC) scattering, plays a crucial role in generating high-energy emission in X-ray and gamma-ray bands.

In the SSC mechanism, synchrotron photons act as seed photons and are upscattered by the same population of relativistic electrons via inverse Compton scattering. In the Thomson regime, the characteristic SSC frequency scales as

$$\nu_{\text{SSC}} \approx \frac{4}{3} \gamma_e^2 \nu_{\text{syn}}, \quad (10)$$

where γ_e is the electron Lorentz factor.

The SSC luminosity is extremely sensitive to relativistic beaming and therefore provides an additional discriminator between blazars and misaligned radio galaxies.

6 Progressive Development of the Simulation

The simulation framework was developed incrementally to isolate and understand the role of each physical ingredient.

1. **Pure Kinematics:** A constant Lorentz factor jet with fixed viewing angle was used to establish baseline Doppler boosting effects.
2. **Curvature Effects:** A spatially varying viewing angle was introduced, producing non-uniform brightness profiles along the jet.

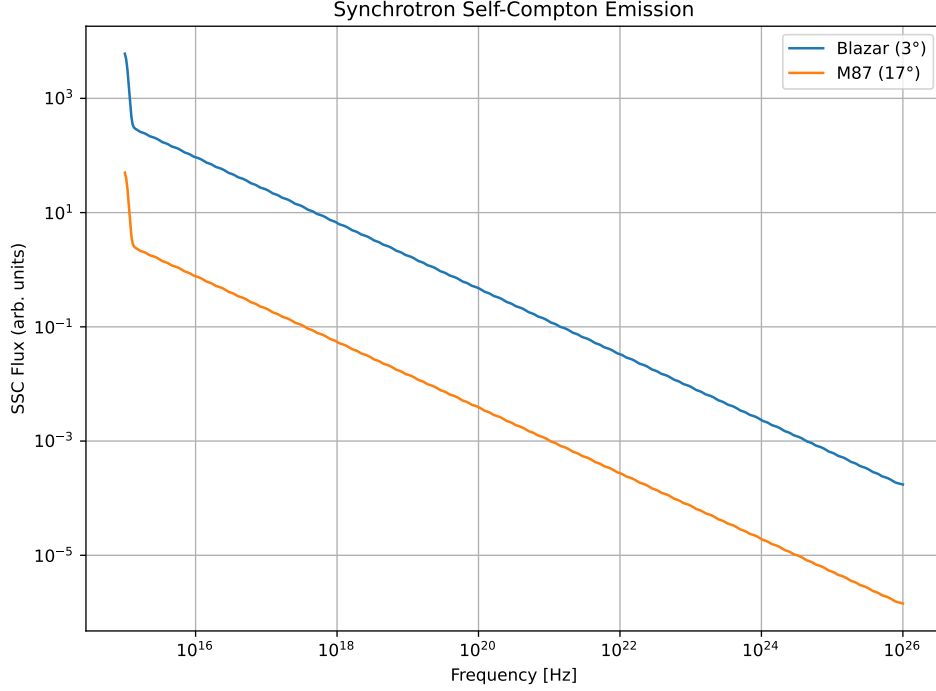


Figure 6: Simulated synchrotron self-Compton (SSC) spectra for a blazar and M87. The high-energy emission is significantly enhanced in the blazar case due to stronger Doppler boosting.

3. **Radiative Physics:** Synchrotron emission with a cooling break was implemented, linking electron energy losses to observable spectral features.
4. **Source Comparison:** Identical intrinsic jet parameters were applied to blazars and M87, demonstrating the unification paradigm.
5. **High-Energy Processes:** SSC emission was added, extending the model to X-ray and gamma-ray regimes.
6. **Time Dependence:** Jet precession generated variability patterns consistent with observed blazar lightcurves.

This modular approach ensures physical transparency and allows straightforward future extensions.

7 Discussion

These simulations demonstrate that:

- Viewing angle alone can explain large differences in observed brightness.
- Synchrotron cooling produces characteristic spectral breaks.
- Jet curvature modifies spatial brightness profiles.
- Relativistic beaming amplifies even small geometric variations into observable variability.

The model is intentionally simplified, but it captures the essential physics governing AGN jet observations.

8 Extended Discussion

The inclusion of SSC emission highlights the multi-zone, multi-process nature of AGN jets. While synchrotron radiation dominates radio-to-optical bands, inverse Compton processes are essential for explaining high-energy observations.

The comparison between blazars and M87 reinforces the AGN unification model: dramatically different observational appearances arise primarily from orientation effects rather than intrinsic differences.

Despite its simplicity, the present model captures the essential physics and provides a robust platform for more advanced treatments, including magnetic field evolution, particle acceleration, and full radiative transfer.

9 Conclusion

We constructed a step-by-step numerical framework to model relativistic jets in AGN. By applying the same intrinsic jet parameters to different viewing angles, we reproduced the qualitative differences between blazars and radio galaxies such as M87. The results highlight the central role of special relativity in high-energy astrophysics and provide a solid foundation for more advanced modeling.

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10 References

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